

(42)



$$\begin{aligned} & 42x^2y^3z \\ & 2, 3, 7, x, y, z \\ & \underline{2 \times 3 \times 7 \times x \times x \times y \times y \times z} \end{aligned}$$

$$\begin{aligned} & 3 \times 3 \times x \times x \\ & 2 \times 3 \times x \times y \\ \text{LCF} &= 3 \times x \\ &= 3x \end{aligned}$$



$$a^2 - b^2 = (a+b)(a-b)$$

$$94^2 - 6^2 = (94+6)(94-6)$$

$$\begin{aligned} \alpha + \beta &= 4k & \alpha + \beta &= 2 \times \alpha \beta \\ (\alpha \beta) &= k+3 & 4k &= 2 \times (k+3) \end{aligned}$$

$$\frac{94^2 - 6^2}{(94-6)} = \frac{(94+6)(94-6)}{(94-6)}$$

$$4k - 2k = 6$$

$$4k = 2k + 6$$

$$18a^{\textcircled{3}}b + 27a^2b$$

$$9a^2b(2a + 3)$$

$$3(6 \quad 9)$$

~~5~~ 7

$$x^2 + xy + xz$$

$$x(x + y + z)$$

$$\text{let 1 diag} = \textcircled{x}$$

$$\text{other} = \textcircled{2x}$$

$$\text{area} = 128$$



$$y^2 - 2y - 7$$

$$x^2 - (x + \beta)x + \alpha\beta$$

$$\checkmark (x + \beta)^3 = x^3 + \beta^3 + 3x\beta(x + \beta)$$

$$x + \beta = 2$$

$$\boxed{\alpha\beta} = -7$$

$$(x + \beta)^2 = x^2 + \beta^2 + 2\boxed{\alpha\beta}$$

$$2^2 = x^2 + \beta^2 + 2 \times (-7)$$

$$4 = x^2 + \beta^2 - 14$$

$$4 + 14 = x^2 + \beta^2$$

$$\boxed{18}$$

$$\frac{2}{\sqrt{2}} = \frac{\sqrt{2} \times \sqrt{2}}{\sqrt{2}} \quad \frac{\sqrt{2}}{2\sqrt{2}}$$

$$\sqrt{2} + \sqrt{2} = 2\sqrt{2}$$

$$\begin{aligned} \text{Angles} &= 1:2:3:4 \\ &= \underline{x, 2x, 3x, 4x} \end{aligned}$$

$$x + 2x + 3x + 4x = 360^\circ \quad \{ \text{sum of } \dots \}$$

$$x = ?$$

$$\begin{aligned} \angle 1 &= 36^\circ \\ \angle 2 &= 2 \times 36 \\ \angle 3 & \\ \angle 4 & \end{aligned}$$

$$n = 55$$

$$S = \frac{n}{2} [2a + (n-1)d]$$

$$3300 = \frac{55}{2} [2a + 54d]$$

$$\frac{3300 \times 2}{55} = 2a + 54d$$

$$60 \times 2 = 2a + 54d$$

$$60 = 2a + 54d$$

$$a + 27d = 60$$

$$27 \times 2 = 54$$

$$\frac{\sin^2 \theta}{\cos \theta} + \cos \theta = \sec \theta$$

$$LHS = \frac{\sin^2 \theta}{\cos \theta} + \frac{\cos \theta}{1}$$

$$k^2 - 144 = 0$$

$$k^2 = 144$$

$$k = \pm 12$$

$$= \frac{\sin^2 \theta}{\cos \theta} + \frac{\cos^2 \theta \times \cos \theta}{1 \times \cos \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta} + \frac{\cos^2 \theta}{\cos \theta}$$

$$= \frac{1}{\cos \theta}$$

$$= \sec \theta = RHS \quad \underline{\underline{HP}}$$

$$(a-d)(a+d) \times a$$

$$(a^2 - d^2) \times a = 504$$

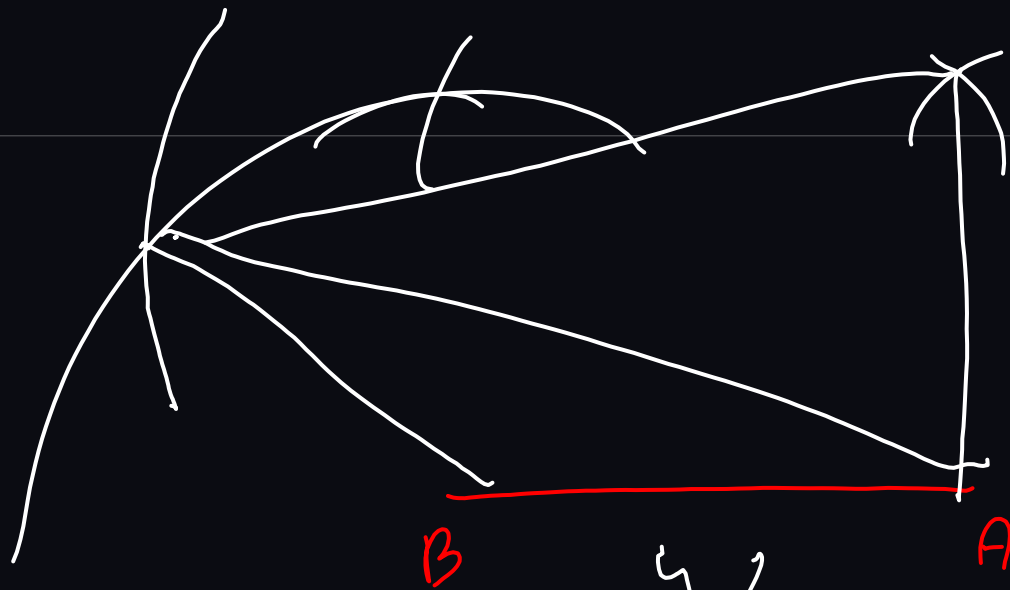
$$(9^2 - d^2) \times 9 = 504$$

$$(81 - d^2) \times 9 = 504$$

$$81 - d^2 = \frac{504}{9}$$

$$81 - d^2 = 56$$

$$d^2 = 81 - 56$$



$$v \cos^2 \theta (1 + \tan^2 \theta) = \underline{1} \quad 4.2$$

$$\begin{aligned} U_1 &= v \cos^2 \theta \times \sec^2 \theta \\ &= \cancel{v \cos^2 \theta} \times \frac{1}{\cancel{v \cos^2 \theta}} \end{aligned}$$

$$LHS = \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}}$$

Rationalize

$$= \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta} \times \frac{(1 - \sin \theta)}{(1 - \sin \theta)}}$$

$$= \sqrt{\frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta}}$$

$$= \sqrt{\frac{(1 - \sin \theta)^2}{\cos^2 \theta}}$$

$$= \frac{1 - \sin \theta}{\cos \theta}$$

$$= \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \sec \theta - \tan \theta$$

$$= RHS \quad \underline{Hr}$$